

Subject: Computer Science Theory**Time allowed:** 2 hours**Maximum marks:** 50**Date:** Practice paperName: _____

Instructions

- All questions are compulsory.
- The marks for each question are shown in brackets on the right.
- Write all your answers in the answer sheet provided, not on this paper.
- Show your working for conversions, truth tables and traces — the working carries marks.

Part A — Multiple Choice Questions*8 questions — 8 marks*

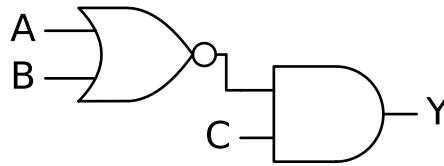
Tick (✓) the correct option for each question.

Q1. What is $1011_2 + 110_2$?

[1]

- (a) 1101_2
 - (b) 10001_2
 - (c) 10101_2
 - (d) 1111_2
-

Q2. Which of the following is the Boolean expression for the output Y of the circuit below? [1]



- (a) $Y = \overline{A} \cdot \overline{B} \cdot C$
 - (b) $Y = \overline{A} + \overline{B} + C$
 - (c) $Y = \overline{A \cdot B} \cdot C$
 - (d) $Y = \overline{A + B + C}$
-

Q3. How many bits are there in an IPv4 address such as 192.168.1.1? [1]

- (a) 8
 - (b) 16
 - (c) 32
 - (d) 64
-

Q4. Cell C2 of a Calc sheet contains =A2*B2. You copy C2 and paste it into C3. What does C3 now contain? [1]

- (a) =A2*B2
 - (b) =A3*B3
 - (c) =A2*B3
 - (d) =C2*C2
-

Q5. What is the value of $7 \% 2 + 7 // 2 * 2$? [1]

- (a) 7
 - (b) 8
 - (c) 6
 - (d) 4
-

Q6. What does $A + A \cdot B$ simplify to?

[1]

- (a) A
 - (b) B
 - (c) $A \cdot B$
 - (d) $A + B$
-

Q7. A picture is 100 pixels wide and 100 pixels tall, and stores one byte for each pixel. About how big is the file?

[1]

- (a) 100 bytes
 - (b) 1 KB
 - (c) 10 KB
 - (d) 100 KB
-

Q8. You press Ctrl + Z once too often. Which shortcut brings back the change you have just undone?

[1]

- (a) Ctrl + Y
 - (b) Ctrl + Z
 - (c) Ctrl + R
 - (d) Ctrl + B
-

Part B — Boolean Algebra

2 questions — 6 marks

Q9. For the Boolean expression below:

[3]

$$Y = A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot B \cdot C$$

- (i) Draw the complete truth table for Y .
 - (ii) Find the value of Y when $A = 1$, $B = 0$, $C = 0$ and when $A = 0$, $B = 1$, $C = 1$. Show the substitution.
-

Q10. For the Boolean expression below:

[3]

$$Y = \overline{(A + B) \cdot C}$$

- (i) Draw the logic-gate circuit for Y .
 - (ii) Draw the truth table for the same circuit.
-

Q11. Draw the logic-gate circuit for the Boolean expression below. [1]

$$Y = \overline{A + B} \cdot C$$

Q12. The shaded cells in the spreadsheet below are selected. Write the address of this selection. [1]

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						

Q13. What is the keyboard shortcut for printing a document in LibreOffice Writer? [1]

Q14. A sound recording stores 1000 samples, and each sample uses 2 **bytes**. How many **bits** is the whole recording? [1]

Q15. What is the value of $(23 \% 5) * (23 // 5)$? [1]

Q16. Which of the following turns a program written by a programmer into the machine code a computer can run? [1]

- (a) A compiler
- (b) An operating system
- (c) A spreadsheet program
- (d) A web browser

Q17. The spreadsheet below is open in LibreOffice Calc. Write what appears in each cell after the formula is typed into it and Enter is pressed. [2]

	A	B	C	D
1	Item	Sales		
2	Pen	120		
3	Book	300		
4	Pen	180		
5	Bag	450		
6	Book	250		
7	Pen	100		
8				
9				

(i) Cell B8 holds the formula =MAX(B2:B7). What does B8 show?

(ii) Cell B9 holds the formula =SUMIF(A2:A7,"Pen",B2:B7). What does B9 show?

Part D — Number Systems

2 questions — 5 marks

Q18. Convert the hexadecimal number $3C7_{16}$ to decimal.

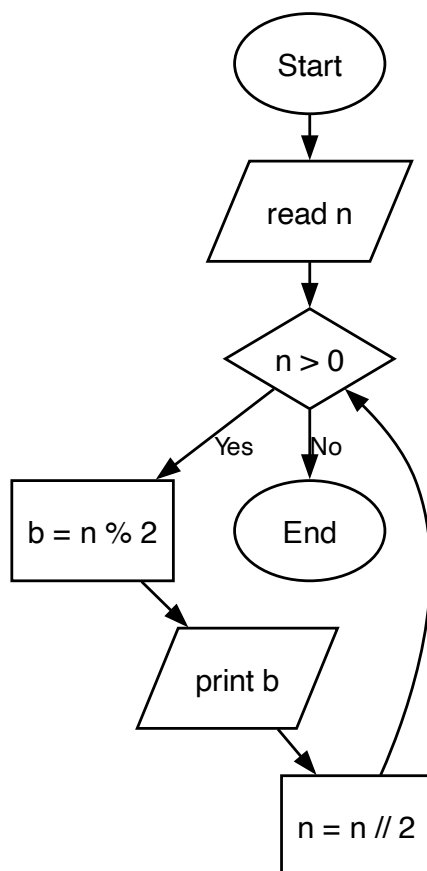
[2]

Q19. Add the binary numbers 101101_2 and 11011_2 . Give your answer in binary.

[3]

Q20. Trace this algorithm for $n = 45$, and draw the trace table.

[5]

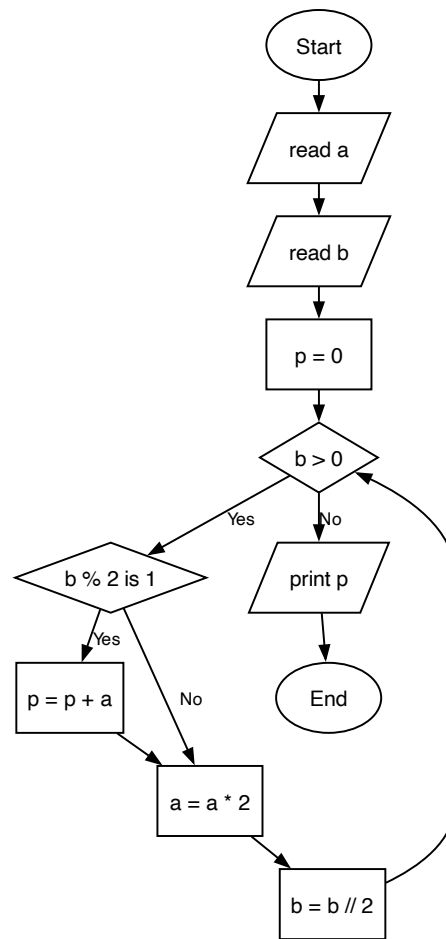


Copy this table into your answer sheet and complete it.

n	b	output

Q21. Trace this algorithm for $a = 13, b = 11$, and draw the trace table.

[4]

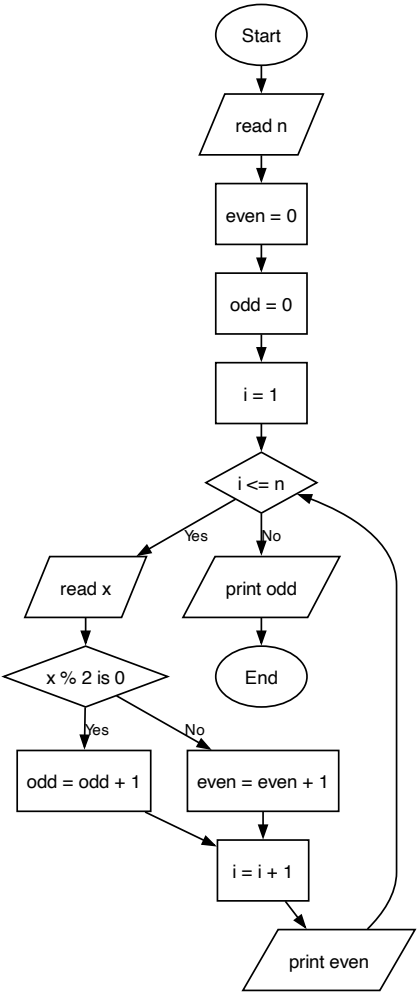


Copy this table into your answer sheet and complete it.

a	b	p	output

Q22. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

The program should read how many numbers there are, then read that many numbers, and finally print how many of them are even and then how many are odd — one number each.

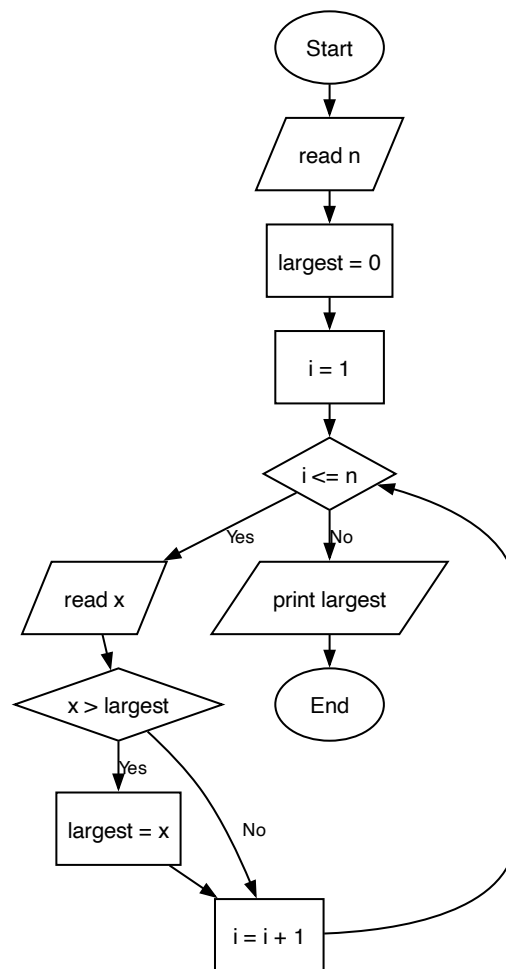


input	should print	actually prints
n = 5, x = 4, 7, 2, 9, 6	3, 2	0, 1, 1, 2, 2, 3

Q23. The flowchart below is **meant** to do what the italic line says, but there are mistakes in it. The table under the flowchart shows what it should print and what it actually prints. **Find all the mistakes in this flowchart, then draw the corrected flowchart in your answer sheet**, so that it gives the correct expected output. [2]

The program should read how many numbers there are, then read that many numbers, and print the largest of them. The numbers may be negative.

The program gives the right answer on one of these test runs and the wrong answer on the other. Say which box is wrong, what it should say, and why the mistake does not show on both runs.



input	should print	actually prints
n = 4, x = 3, 7, 2, 5	7	7
n = 4, x = -5, -2, -9, -1	-1	0

Q24. Read the description below, then answer the parts that follow.

[7]

A program reads a whole number n and prints the sum of its digits. For example, if n is 302 the program prints 5, because $3 + 0 + 2 = 5$. (Hint: $n \% 10$ is the last digit of n , and $n // 10$ is n with that digit removed.)

- (i) Write the algorithm in words, step by step.
 - (ii) Draw the flowchart for that algorithm.
 - (iii) Trace your flowchart for $n = 4728$ and $n = 999$, and write what it prints in each case.
-

Q25. Read the description below, then answer the parts that follow.

[3]

*A factory works out a worker's weekly pay. The program reads the number of hours worked. Every hour is paid at Rs 50. If more than 40 hours were worked, each hour *above* 40 is paid a second time over, at another Rs 50 an hour. The program then prints the total pay.*

- (i) Write the algorithm in words, step by step.
- (ii) Draw the flowchart for that algorithm.